

Testing of Hypothesis: Introduction

Reading Objectives:

In this reading, you will learn about sampling and its importance in various applications. You will be introduced to sampling distribution and also various distributions associated with it, such as t-distribution and chi-square distribution.

Main Reading Section:

Introduction to testing of hypothesis

In general, we wish to test a claim made with respect to population. Testing of hypothesis is very important in the validation of inferences or hypotheses related to population.

For example, “Smoking causes cancer” is a hypothesis (or assumption) we want to validate based on the relevant data and sampling observations.

Another example is suppose that we want to validate the hypothesis that after the recent pandemic, most people prefer over-the-top (OTT) platforms to watch movies rather than watching movies in theaters. We need to collect a sample consisting of people who watch movies in theaters/OTT. Based on the sample analysis using suitable statistical tests, we need to validate the hypothesis.

Null and Alternative Hypotheses

Let us consider that we want to check that there is

No difference in the effectiveness of the two teaching methods

No difference in the cost of the two methods of irrigation

No difference in the yield strength of the two kinds of steel

We call these hypotheses null hypotheses and denote them by H_0 .

Formulate a null hypothesis and an appropriate alternative hypothesis that we accept when the null hypothesis must be rejected.

Consider the case where we want to test the hypothesis, i.e., when the null hypothesis is $\mu = 20$ minutes and the alternative hypothesis is $\mu > 20$ minutes. This kind of alternative is called a **one-sided alternative or a right-sided alternative**.

Consider the case where we want to test the hypothesis, i.e., when null hypothesis is $\mu = 20$ minutes and the alternative hypothesis is $\mu < 20$ minutes. This kind of alternative is called a **one-sided alternative or a left-sided alternative**.

Consider the case where we want to test the hypothesis, i.e., when null hypothesis is $\mu = 20$ minutes and the alternative hypothesis is $\mu \neq 20$ minutes. This kind of alternative is called a **two-sided (two-tailed) alternative**.

These help us to decide the critical region in case of z-test/t-test.

Example:

A food processor wants to check whether the average amount of coffee that goes into his 4-ounce jars is indeed 4 ounces.

Since the food processor cannot afford to put much less than 4 ounces into each jar for fear of losing customer acceptance, nor can he afford to put much more than 4 ounces into each jar for fear of losing part of his profit, the appropriate alternative hypothesis is $\mu \neq 4$.

Type I and Type II Errors

In hypothesis testing, there are two types of errors. **Type I error:** A type I error occurs when we incorrectly reject H_0 (i.e., we reject the null hypothesis, when H_0 is true). **Type II error:** A type II error occurs when we incorrectly fail to reject H_0 (i.e., we accept H_0 when it is not true).

Decision	Observation	
	H_0 is true	H_0 is false
H_0 is accepted	Decision is correct	Type II error
H_0 is rejected	Type I error	Decision is correct